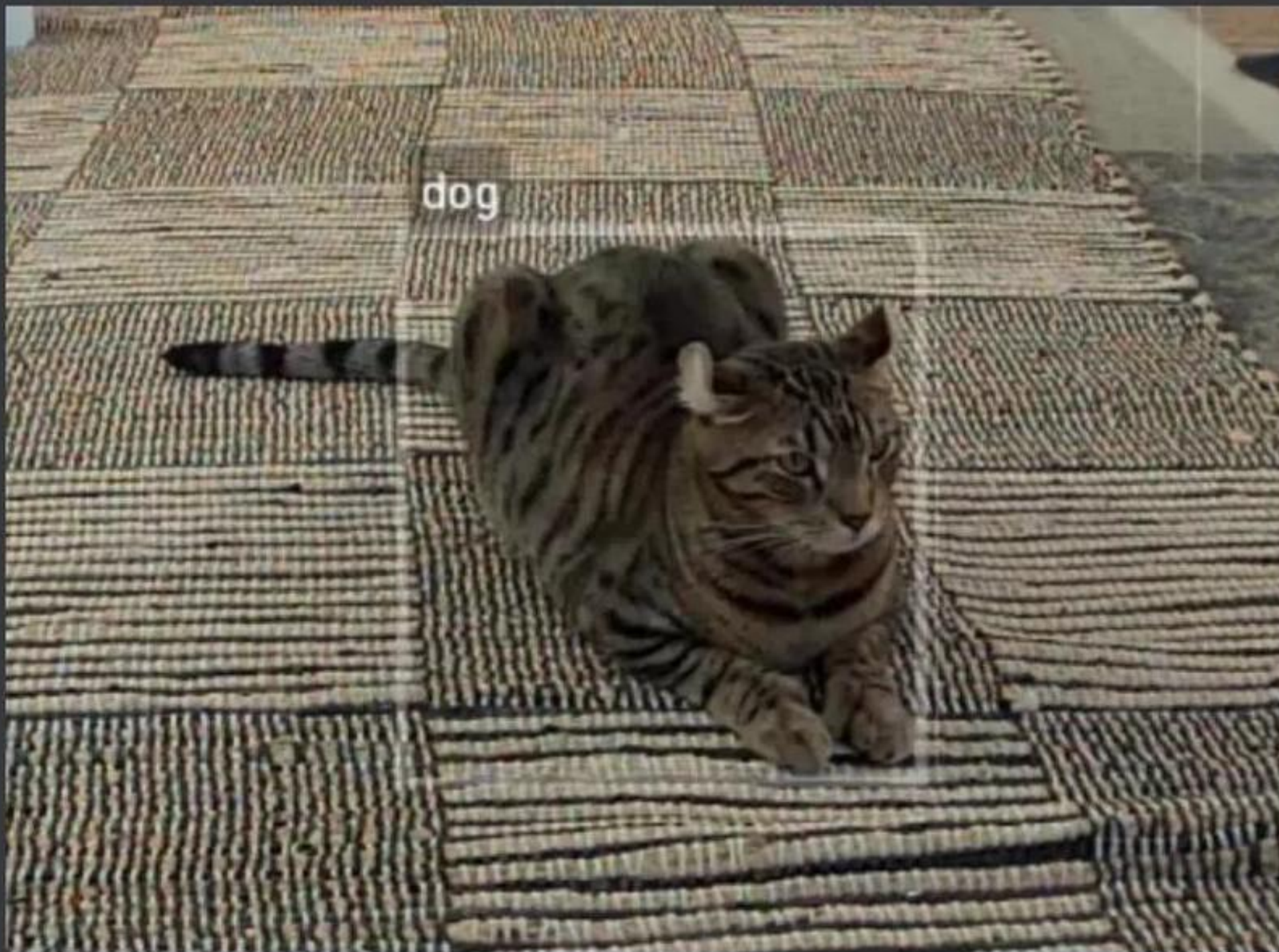
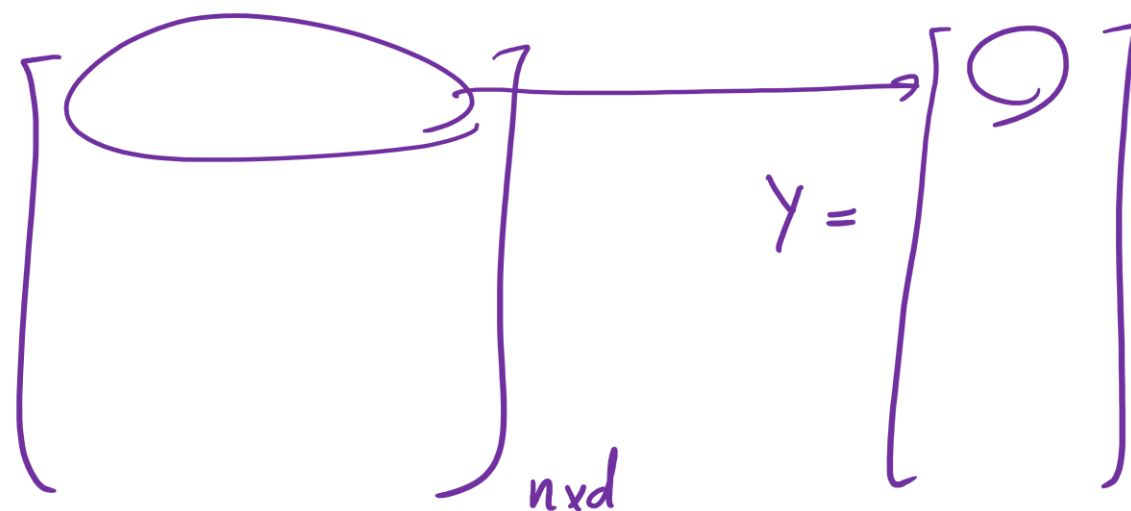
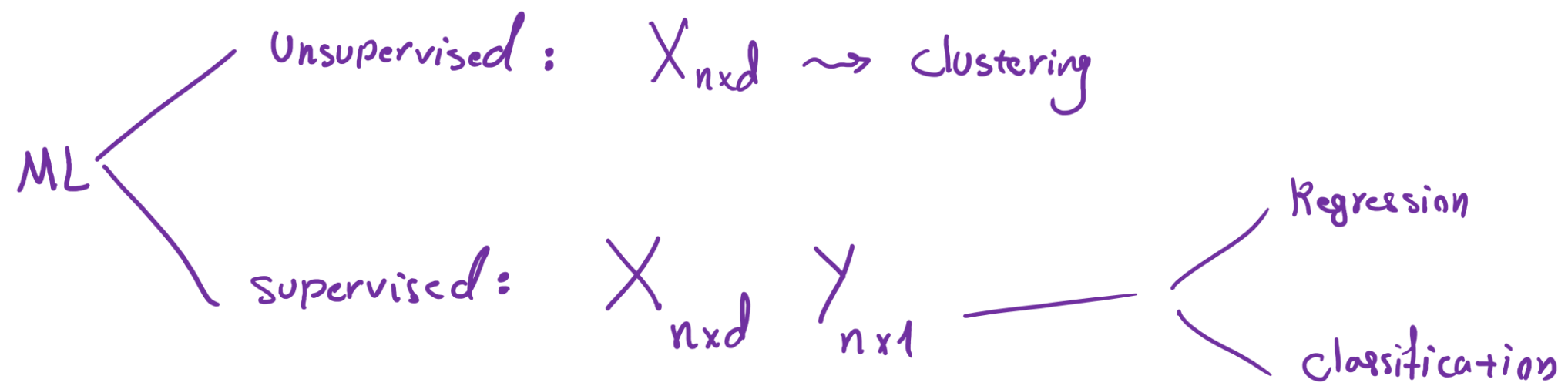


Clustering Analysis and K-Means

Mahdi Roozbahani
Georgia Tech

60+ hours on 16 GPU nvidia CUDA cluster.

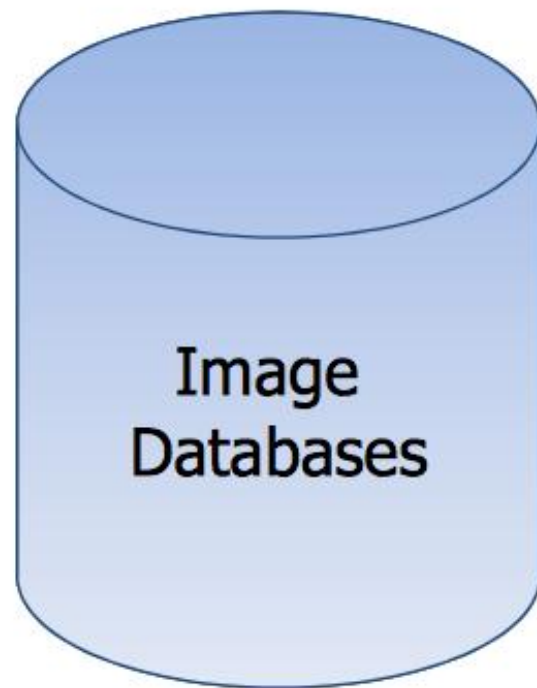




Outline

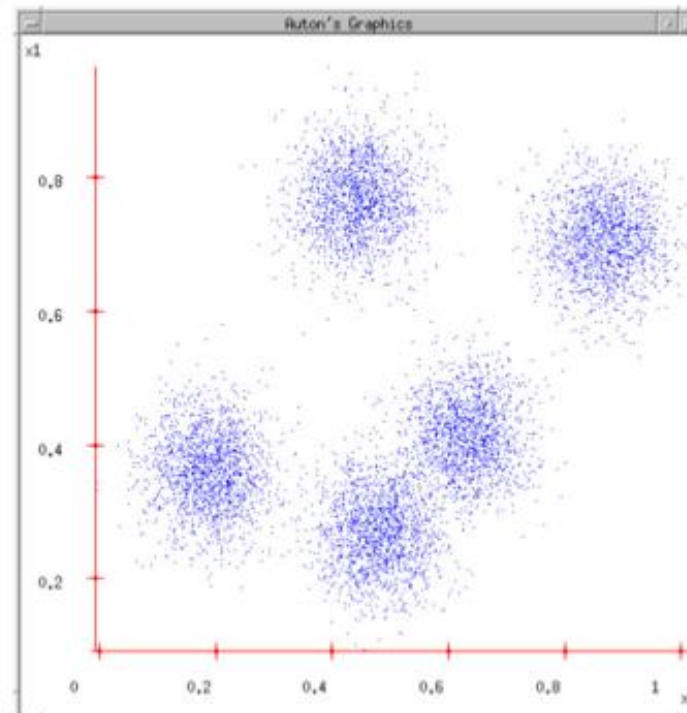
- Clustering 
- Distance Function
- K-Means Algorithm
- Analysis of K-Means

Clustering Images



Goal of clustering:

Divide object into groups,
and objects within a group
are more similar than
those outside the group

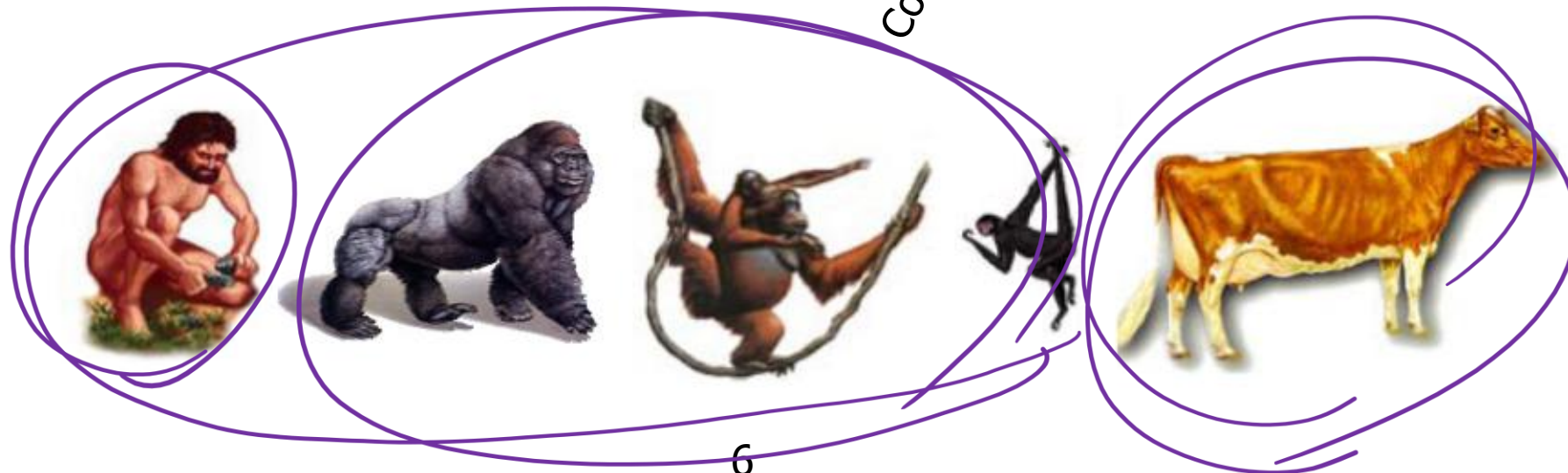


Clustering Other Objects



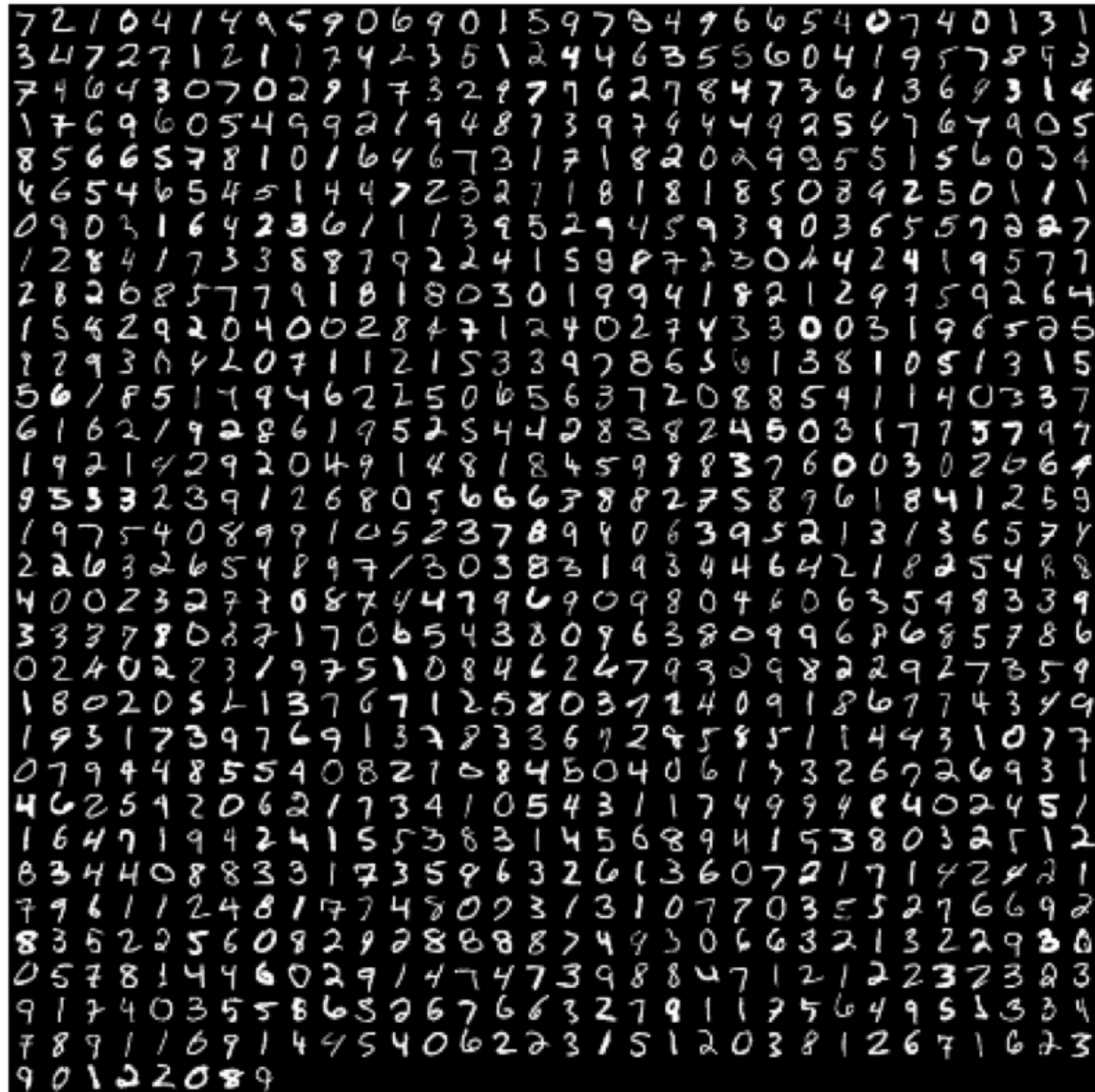
Belarusian **Piotr**
 Azerbaijani **Pyotr**
 Greek **Petros**
 Italian **Pietro**
 Portuguese **Pedro**
 French **Pierre**
 Italian **Piero**
 Dutch **Peter**
 Danish **Peder**
 Couldn't find it – Finish? Peka
 Irish Peadar

Linguistic Similarity

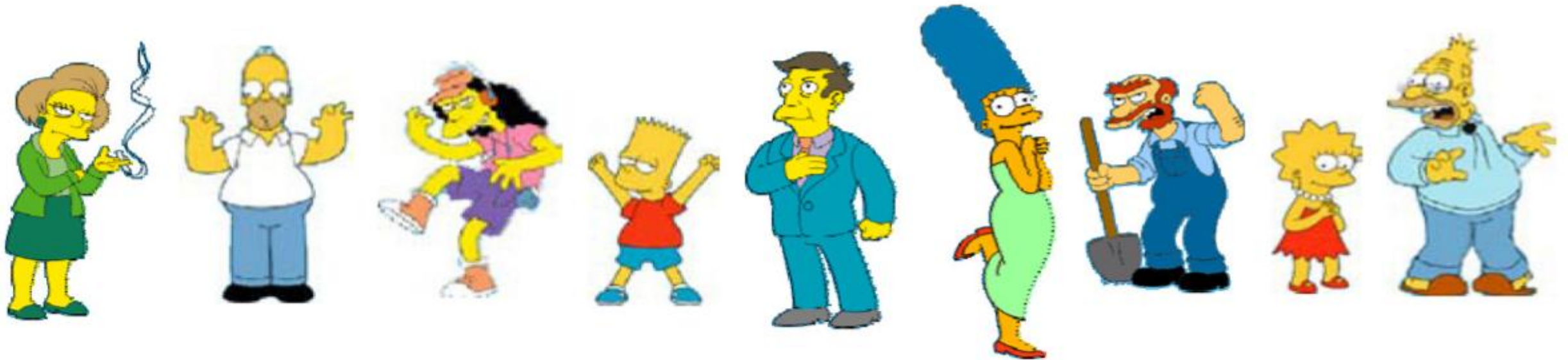


Clustering Hand Digits

0 1 2 3 4 5 6 7 8 9

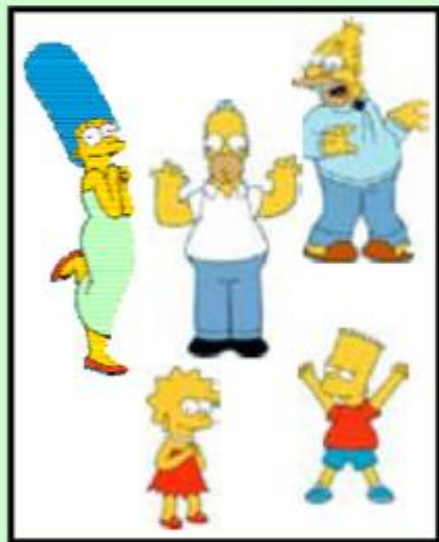


Clustering is Subjective



What is consider similar/dissimilar?

Clustering is subjective



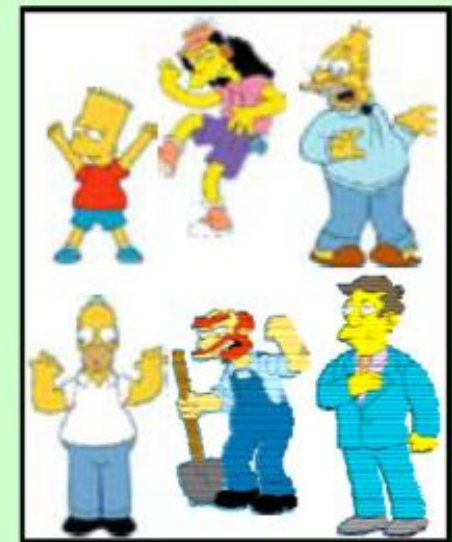
Simpson's Family



School Employees



Females



Males

Are they similar or not?



So What is Clustering in General?

- You pick your similarity/dissimilarity function
- The algorithm figures out the grouping of objects based on the chosen similarity/dissimilarity function
 - Points within a cluster is similar
 - Points across clusters are not so similar
- Issues for clustering
 - How to represent objects? (Vector space? Normalization?)
 - What is a similarity/dissimilarity function for your data?
 - What are the algorithm steps?

Outline

- Clustering
- Distance Function 
- K-Means Algorithm
- Analysis of K-Means

Properties of Similarity Function

- Desired properties of dissimilarity function
 - Symmetry: $d(x, y) = d(y, x)$
 - *Otherwise you could claim "Alex looks like Bob, but Bob looks nothing like Alex"*
 - Positive separability: $d(x, y) = 0$, if and only if $x = y$
 - *Otherwise there are objects that are different, but you cannot tell apart*
 - Triangular inequality: $d(x, y) \leq d(x, z) + d(z, y)$
 - *Otherwise you could claim "Alex is very like Bob, and Alex is very like Carl, but Bob is very unlike Carl"*

Distance Functions for Vectors

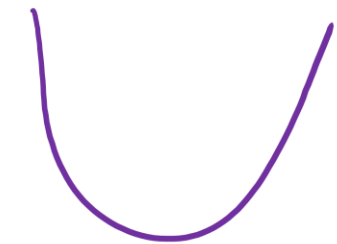
- Suppose two data points, both in R^d

- $x = (x_1, x_2, \dots, x_d)$

- $y = (y_1, y_2, \dots, y_d)$



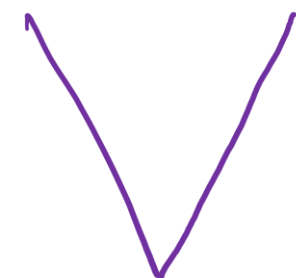
- Euclidean distance: $d(x, y) = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$



- Minkowski distance: $d(x, y) = \sqrt[p]{\sum_{i=1}^d (x_i - y_i)^p}$

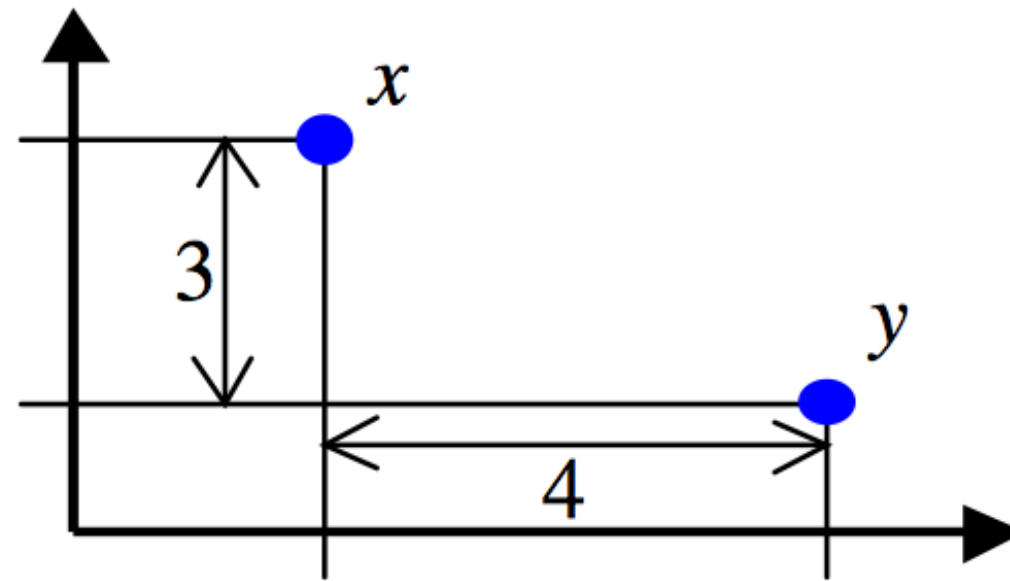
- Euclidean distance: $p = 2$

- Manhattan distance: $p = 1, d(x, y) = \sum_{i=1}^d |x_i - y_i|$



- “inf”-distance: $p = \infty, d(x, y) = \max_{i=1}^d |x_i - y_i|$

Example



- Euclidean distance: $\sqrt{4^2 + 3^2} = 5$
- Manhattan distance: $4 + 3 = 7$
- “inf”-distance: $\max\{4, 3\} = 4$

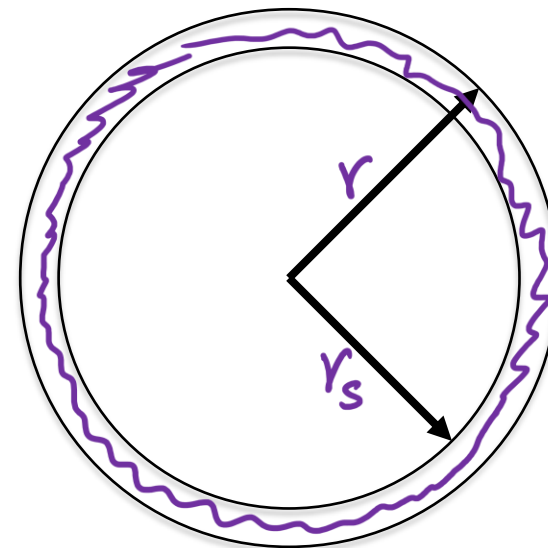
Some problems with Euclidean distance



$$V_T = \frac{4}{3} \pi r^d = c r^d$$

$$V_{\text{shell}} = c r^d - c r_s^d$$

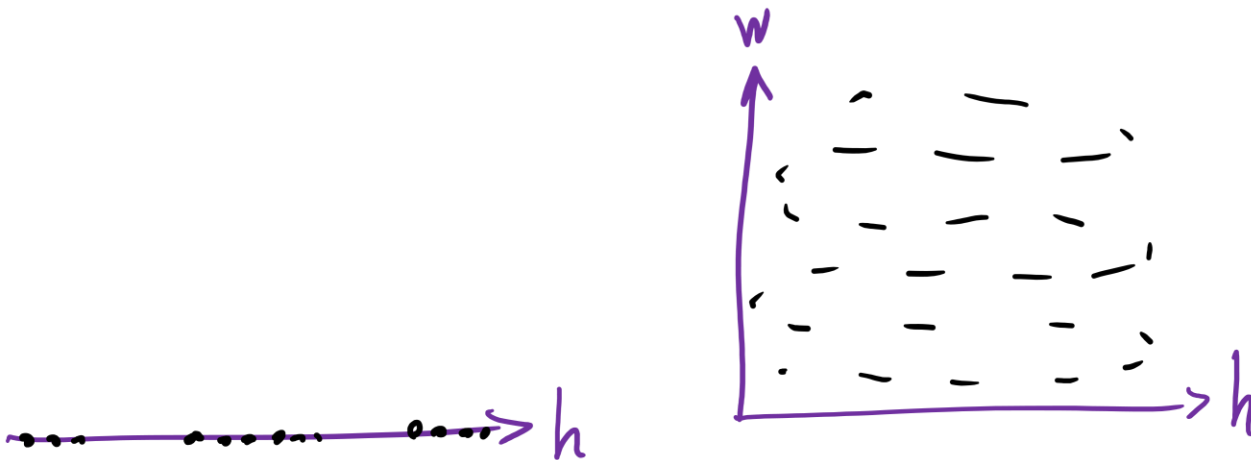
$$\frac{V_{\text{shell}}}{V_T} = \frac{c r^d - c r_s^d}{c r^d} = 1 - \left(\frac{r_s}{r}\right)^d$$



$$0 < \frac{r_s}{r} < 1$$

$$d \rightarrow \infty$$

$$\frac{V_{\text{shell}}}{V_T} \approx 1 \Rightarrow V_{\text{shell}} \approx V_T$$



Hamming Distance

- Manhattan distance is also called *Hamming distance* when all features are binary
 - Count the number of difference between two binary vectors
 - Example, $x, y \in \{0,1\}^{17}$

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
x	0	1	1	0	0	1	0	0	1	0	0	1	1	1	0	0	1
y	0	1	1	1	0	0	0	0	1	1	1	1	1	1	0	1	1

$$d(x, y) = 5$$

Edit Distance

- Transform one of the objects into the other, and measure how much effort it takes

x	I	N	T	E	*	N	T	I	O	N
y	*	E	X	E	C	U	T	I	O	N
	d	s	s		i	s				

d: deletion (cost 5)

s: substitution (cost 1)

i: insertion (cost 2)

$$d(x, y) = 5 \times 1 + 3 \times 1 + 1 \times 2 = 10$$

Outline

- Clustering
- Distance Function
- K-Means Algorithm 
- Analysis of K-Means

Results of K-Means Clustering:



Image



Clusters on intensity



Clusters on color

K-means clustering using intensity alone and color alone



Image



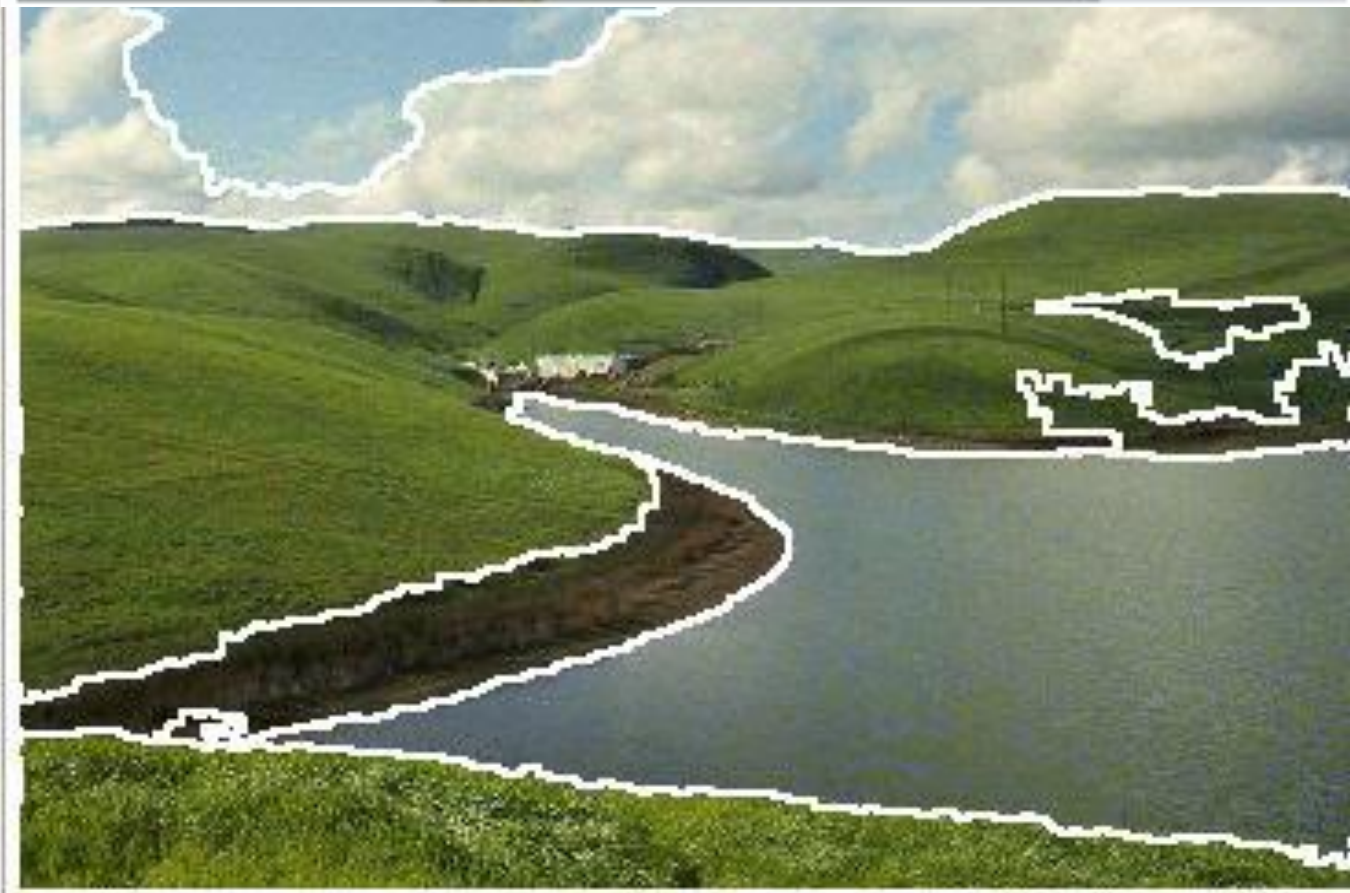
Clusters on color

K-means using color alone, 11 segments (clusters)

①



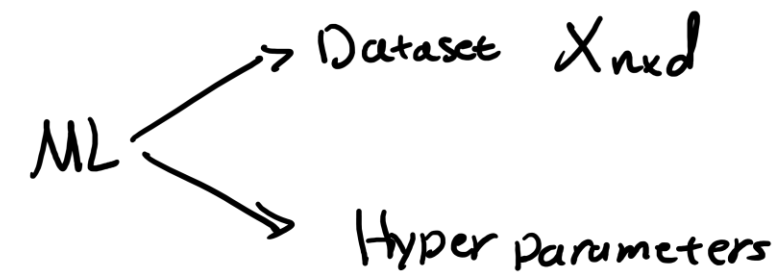
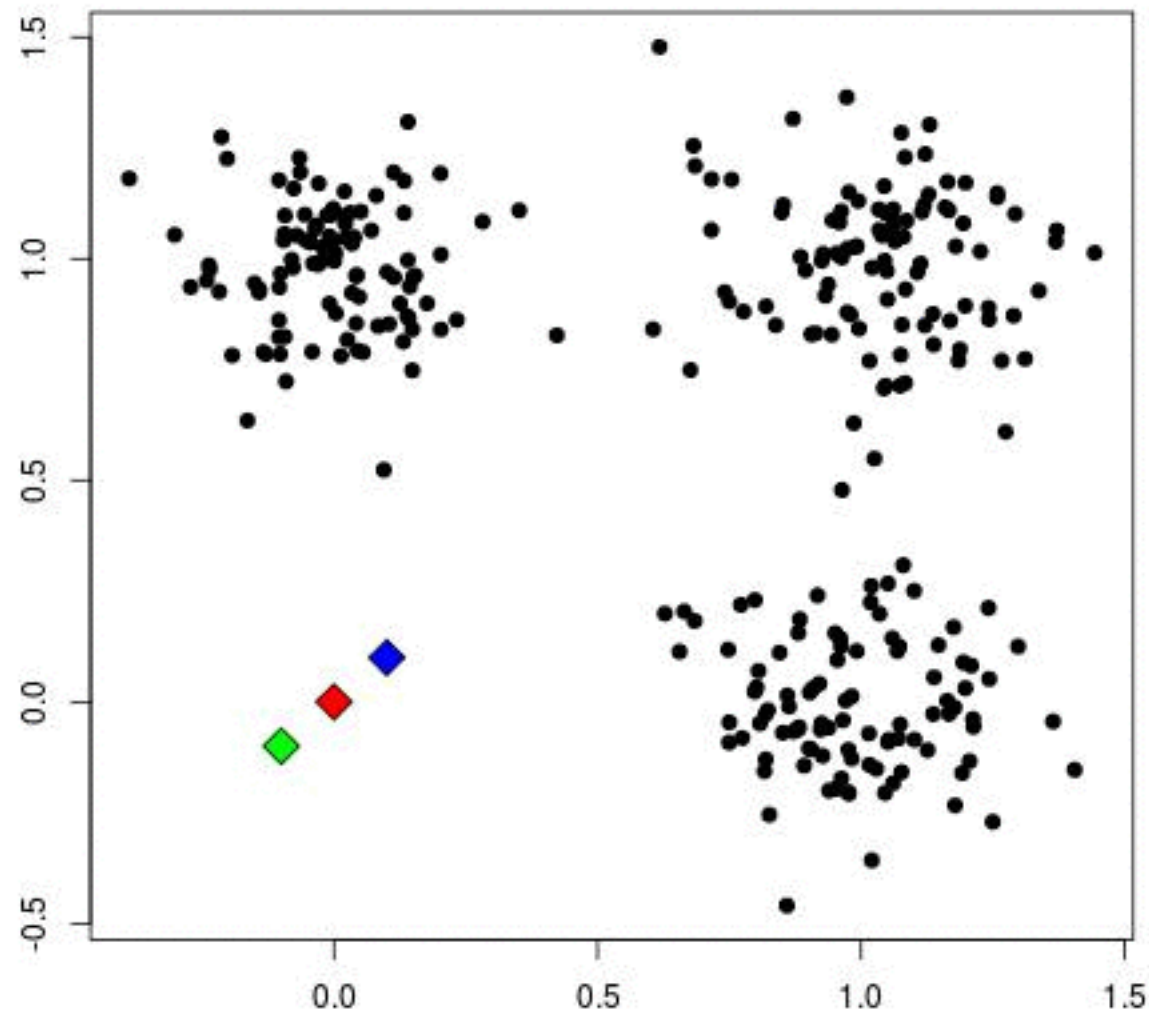
②



K-Means Algorithm

hyper parameter

Start!



Visualizing K-Means Clustering



$$x^{(i)} = \begin{bmatrix} h & w \end{bmatrix}$$

Step 0 : randomly initialize means

Step 1 : membership

Step 2 : re-calculate centers

Step 3 : calculate distortion

$$= a_1 + a_2 + a_3 + b_1 + b_2 + b_3 + b_4$$

K-Means Algorithm

- Initialize k cluster centers, $\{c_1, c_2, \dots, c_k\}$, randomly

- Do

- Decide the cluster memberships of each data point, x_i by assigning it to the nearest cluster center (**cluster assignment**)

$$\pi(i) = \operatorname{argmin}_{j=1,\dots,k} \|x_i - c_j\|^2 \rightsquigarrow \underline{\text{Expectation}}$$

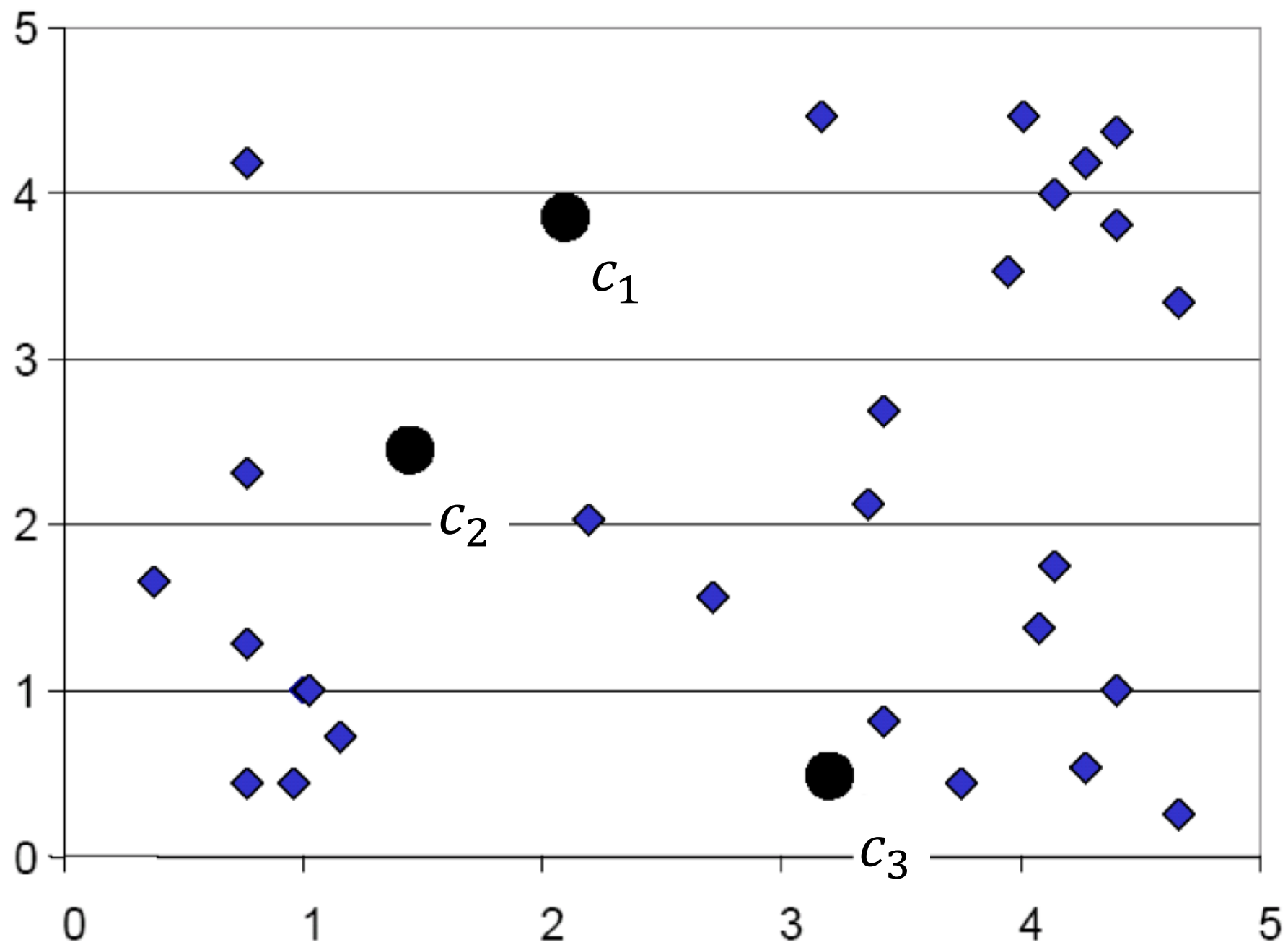
- Adjust the cluster centers (**center adjustment**)

EM

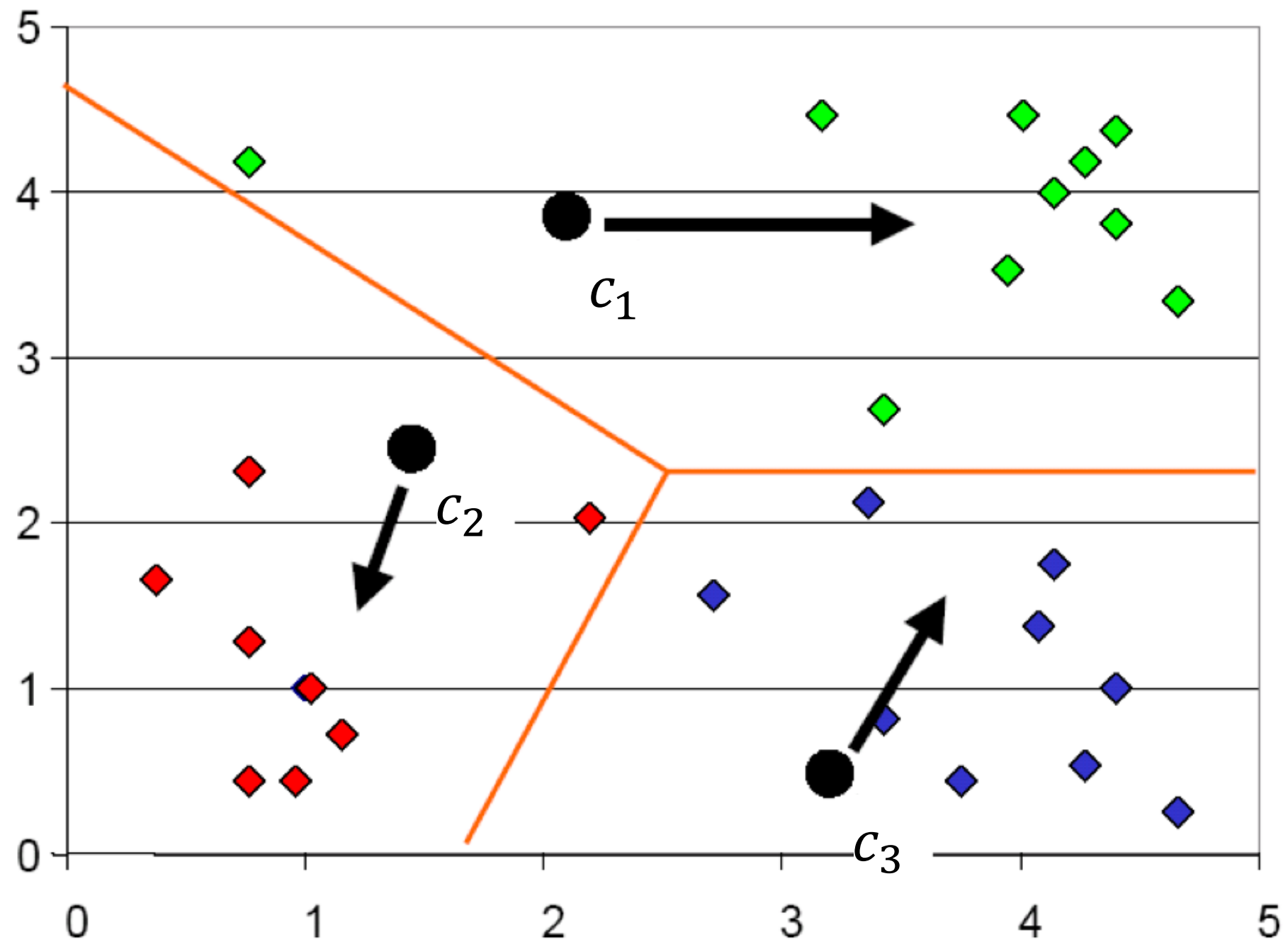
$$c_j = \frac{1}{|\{i: \pi(i) = j\}|} \sum_{i: \pi(i)} x_i \rightsquigarrow \underline{\text{Maximization}}$$

- While any cluster center has been changed

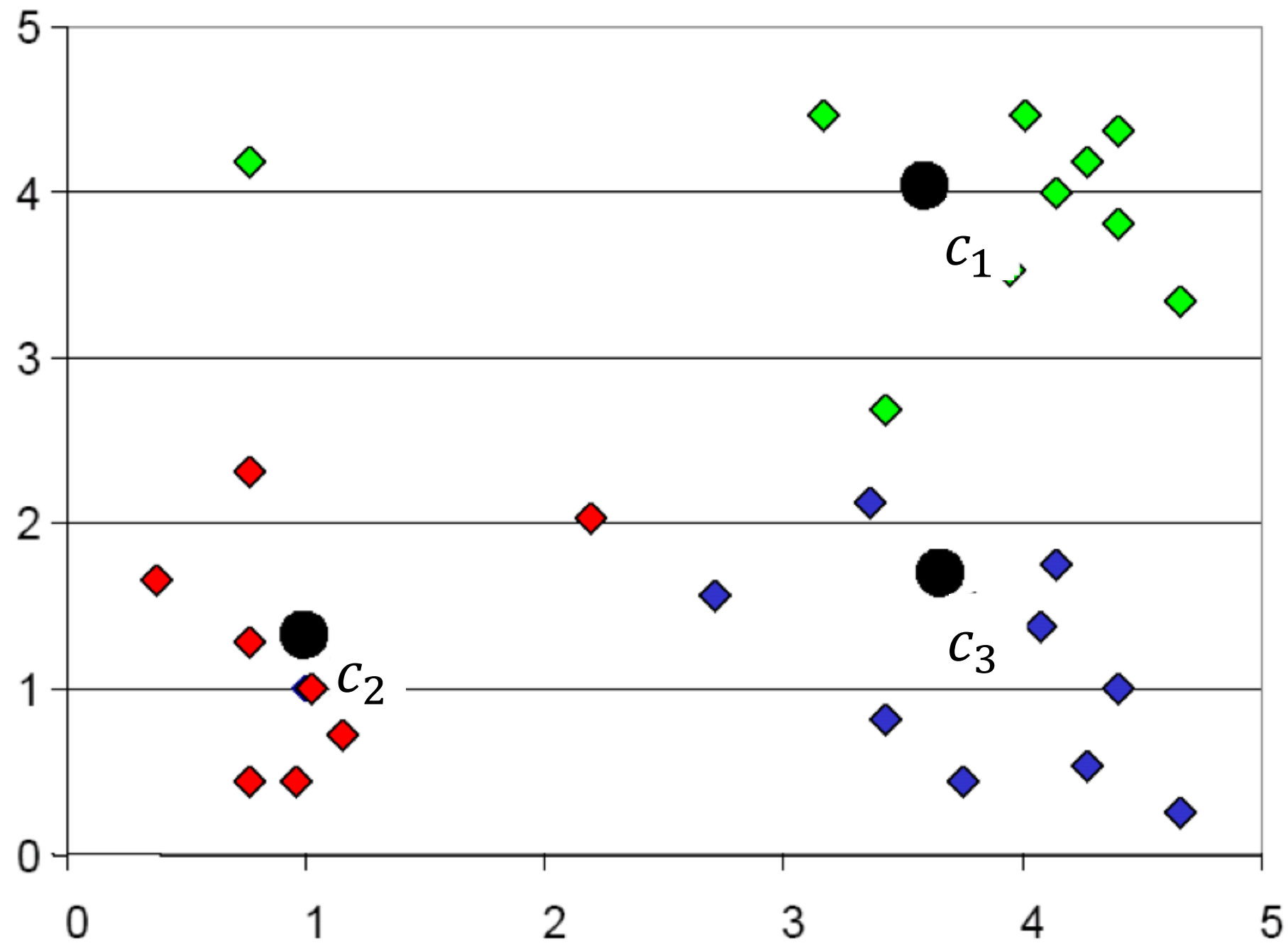
K-Means: Step 1



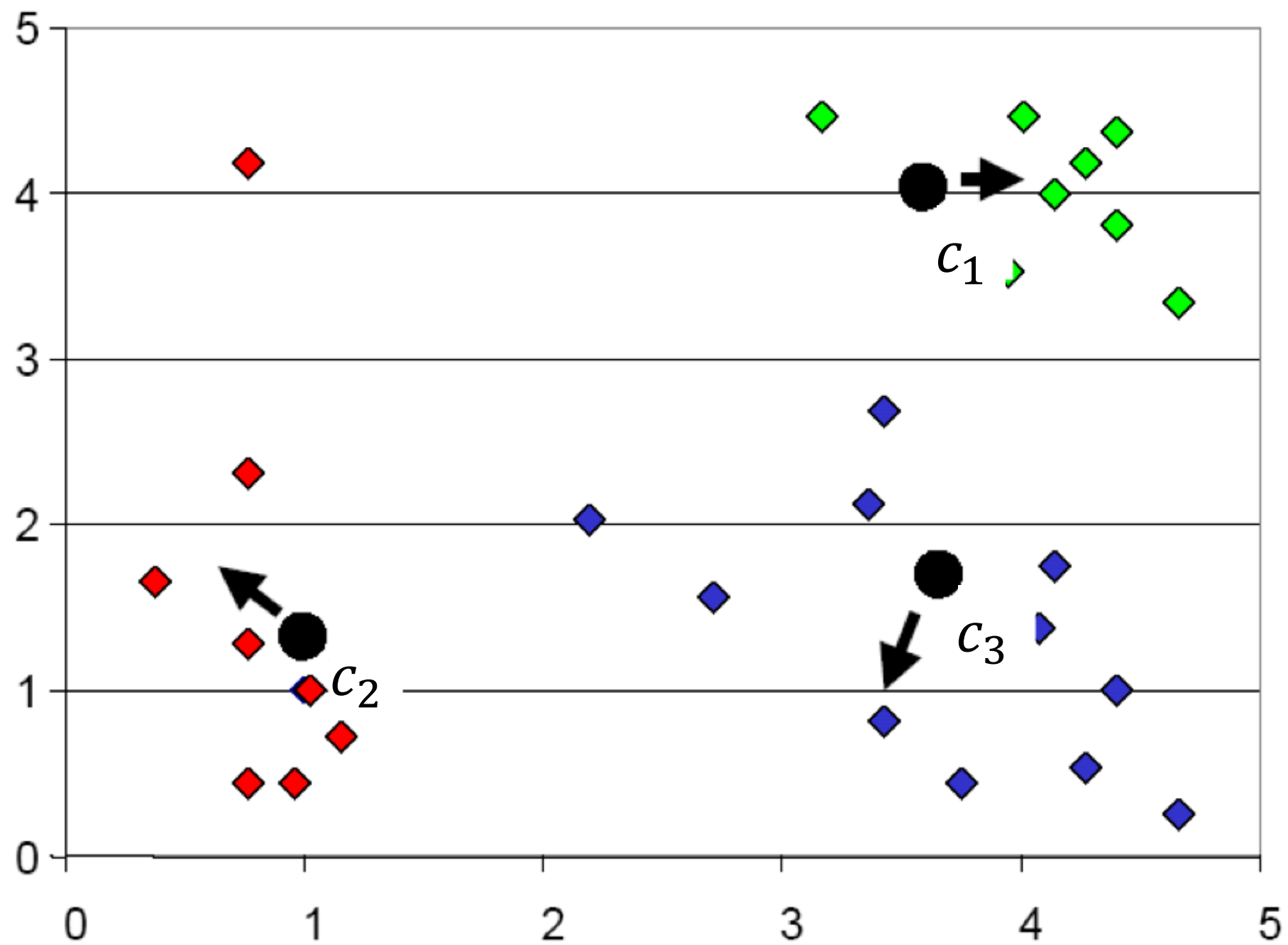
K-Means: Step 2



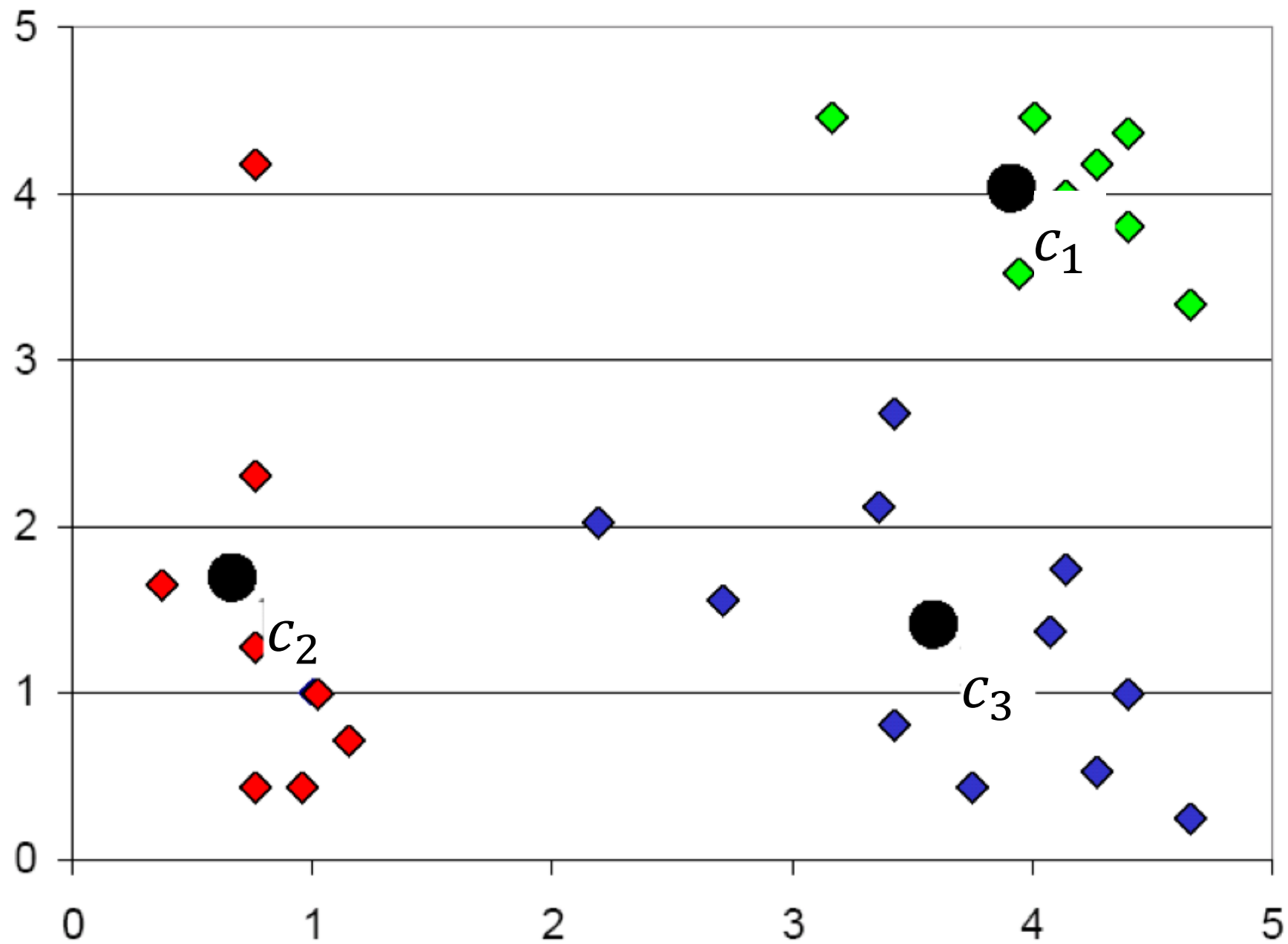
K-Means: Step 3



K-Means: Step 4



K-Means: Step 5



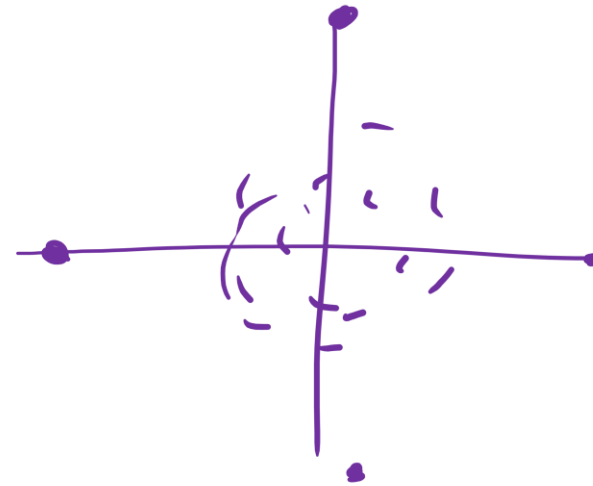
Outline

- Clustering
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Questions

- Will different initialization lead to different results?

- Yes
- No
- Sometimes

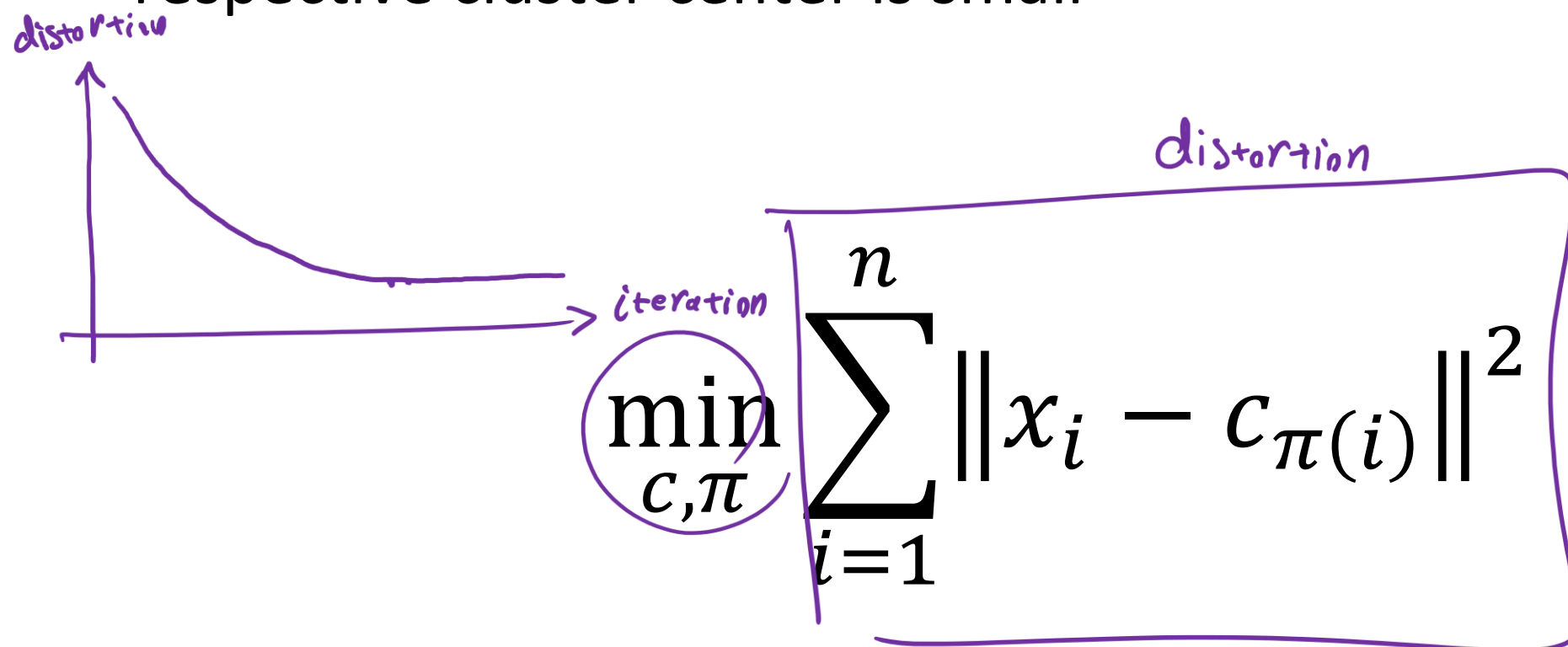


- Will the algorithm always stop after some iteration?

- Yes
- No (we have to set a maximum number of iterations)
- Sometimes

Formal Statement of the Clustering Problem

- Given n data points, $\{x_1, x_2, \dots, x_n\}$ $x \in R^d$
- Find k cluster centers, $\{c_1, c_2, \dots, c_k\}$ $c \in R^d$
- And assign each datapoint i to one cluster, $\pi(i) \in \{1, \dots, k\}$
- Such that the averaged square distances from each datapoint to its respective cluster center is small



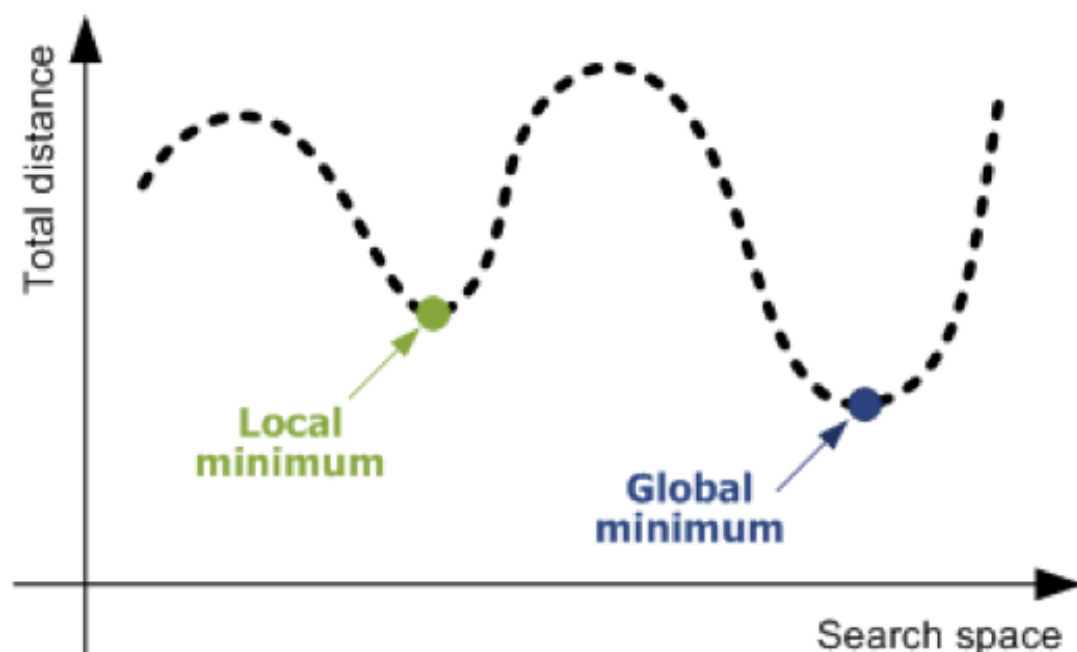
Clustering is NP-Hard

- Find k cluster centers, $\{c_1, c_2, \dots, c_k\} \subset \mathbb{R}^d$, and assign each data point i to one cluster, $\pi(i) \in \{1, \dots, k\}$, to minimize

$$\min_{C, \pi} \sum_{i=1}^n \|x_i - c_{\pi(i)}\|^2$$

NP-hard!

- A search problem over the space of discrete assignments
 - For all n data point together, there are k^n possibility
 - The cluster assignment determines cluster centers, and vice versa



- For all n data point together, there are k^n possibility

$$k^n = 2^3 = 8$$

$$X = \{A, B, C\}$$

$n=3$ (data points)

$k=2$ clusters of two members

Cluster 1

A, B, C

$\{\}$

A

B

C

B, C

A, C

A, B

Cluster 2

$\{\}$

A, B, C

B, C

A, C

A, B

A

B

C

Convergence of K-Means

- Will kmeans objective oscillate?

$$\min_{c, \pi} \sum_{i=1}^n \|x_i - c_{\pi(i)}\|^2$$

- The minimum value of the objective is finite
- Each iteration of kmeans algorithm decrease the objective
 - Cluster assignment step decreases objective
 - $\pi(i) = \operatorname{argmin}_{j=1, \dots, k} \|x_i - c_{\pi(j)}\|^2$ for each data point i
 - Center adjustment step decreases objective
 - $c_i = \frac{1}{|\{i: \pi(i)=j\}|} \sum_{i: \pi(i)=j} x_i = \operatorname{argmin}_c \sum_{i: \pi(i)=j} \|x_i - c_{\pi(j)}\|^2$

Time Complexity

$$X_{n \times d}$$

$$O(\cdot)$$

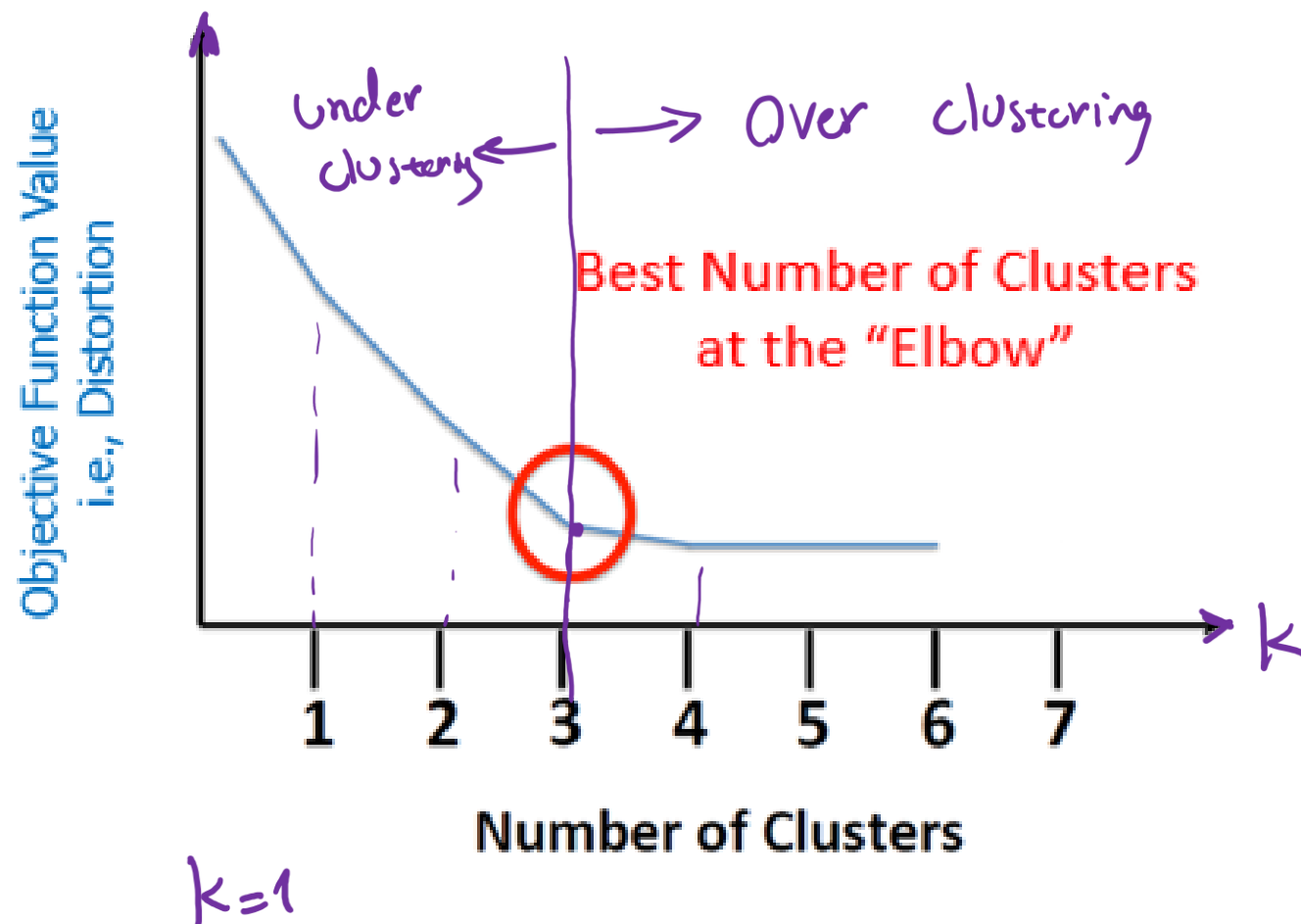
$$\begin{matrix} x^{\{1\}} \\ x^{\{2\}} \end{matrix} = \begin{bmatrix} \text{h} & \text{w} & \text{a} & \dots \\ \text{---} & \text{---} & \text{---} & \text{---} \end{bmatrix}_{1 \times d}$$
$$\|x^{\{1\}} - x^{\{2\}}\|_2^2$$

- Assume computing distance between two instances is $O(d)$ where d is the dimensionality of the vectors.
- Reassigning clusters for all datapoints:
 - $O(kn)$ distance computations (when there is one feature)
 - $O(knd)$ (when there is d features)
- Computing centroids: Each instance vector gets added once to some centroid (Finding centroid for each feature): $O(nd)$.
- Assume these two steps are each done once for I iterations: $O(Iknd)$.

$$\begin{matrix} O(d) \\ \downarrow \\ O(nd) \\ \downarrow \\ O(knd) \\ \downarrow \\ O(Iknd) \end{matrix}$$

How to Choose K?

Elbow method



Distortion score: computing the sum of squared distances from each point to its assigned center